

Basis Data Part10

PENGAMPU:

KHAMARUDIN SYARIF

Query Having

HAVING digunakan untuk menggantikan WHERE ketika menggunakan Group BY yang datanya di agregasi.

Secara umum HAVING digunakan setelah melakukan GROUP BY

Berikut sintaks yang digunakan:

```
SELECT nama_kolom  
FROM nama_table  
GROUP BY nama_kolom  
HAVING kondisi
```



Query Having

Code Editor

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1 Hint

```
1 SELECT customer_id FROM Subscription GROUP BY customer_id HAVING  
COUNT(customer_id) > 1;
```

<u>customer id</u>
1
7
12
20
31
40

Query Having

```
SELECT customer_id, product_id, subscription_date FROM Subscription  
WHERE customer_id IN (SELECT customer_id FROM Subscription  
GROUP BY customer_id HAVING COUNT(customer_id) > 1)  
ORDER BY customer_id ASC;
```

<u>customer_id</u>	<u>product_id</u>	<u>subscription_date</u>
1	10001	2018-10-05 00:00:00
1	10002	2018-12-03 00:00:00
7	10002	2018-10-11 00:00:00
7	10003	2018-12-30 00:00:00
12	10002	2018-10-16 00:00:00
12	10004	2018-12-20 00:00:00
20	10002	2018-10-24 00:00:00
20	10002	2018-12-23 00:00:00
31	10003	2018-10-27 00:00:00
31	10004	2018-12-15 00:00:00
40	10002	2018-10-29 00:00:00
40	10003	2018-12-02 00:00:00

Query Having

```
SELECT b.name, b.address, b.phone, a.product_id, a.subscription_date
FROM subscription a JOIN customer b ON a.customer_id = b.id WHERE b.id IN
(SELECT customer_id FROM Subscription GROUP BY customer_id HAVING
COUNT(customer_id) > 1) ORDER BY b.id ASC;
```

name	address	phone	product_id	subscription_date
Wani Nurdiyanti	Psr. Jayawijaya No. 913, Sawahlunto 10252, SulTeng	(+62) 976 8680 7043	10001	2018-10-05 00:00:00
Wani Nurdiyanti	Psr. Jayawijaya No. 913, Sawahlunto 10252, SulTeng	(+62) 976 8680 7043	10002	2018-12-03 00:00:00
Tugiman Jailani M.Ak	Gg. Jayawijaya No. 429, Tual 23392, SulBar	0862 4695 650	10002	2018-10-11 00:00:00
Tugiman Jailani M.Ak	Gg. Jayawijaya No. 429, Tual 23392, SulBar	0862 4695 650	10003	2018-12-30 00:00:00
Laswi Saputra	Dk. Raden Saleh No. 257, Bandar Lampung 13181, SulTeng	(+62) 899 1181 4688	10002	2018-10-16 00:00:00
Laswi Saputra	Dk. Raden Saleh No. 257, Bandar Lampung 13181, SulTeng	(+62) 899 1181 4688	10004	2018-12-20 00:00:00
Rama Putra	Ds. Abang No. 652, Bengkulu 70019, Bali	0917 8935 641	10002	2018-10-24 00:00:00
Rama Putra	Ds. Abang No. 652, Bengkulu 70019, Bali	0917 8935 641	10002	2018-12-23 00:00:00
Aisyah Umi Agustina	Jr. Astana Anyar No. 416, Batu 57453, SulBar	0372 4694 6834	10003	2018-10-27 00:00:00
Aisyah Umi Agustina	Jr. Astana Anyar No. 416, Batu 57453, SulBar	0372 4694 6834	10004	2018-12-15 00:00:00
Makara Pangestu	Gg. Bank Dagang Negara No. 316, Makassar 90333, Bengkulu	(+62) 946 3174 093	10002	2018-10-29 00:00:00
Makara Pangestu	Gg. Bank Dagang Negara No. 316, Makassar 90333, Bengkulu	(+62) 946 3174 093	10003	2018-12-02 00:00:00



Fungsi Max pada Having

```
Code Editor Run Submit 2 Hint ...  
1 SELECT product_id, MAX(total_price) AS total FROM invoice GROUP BY product_id;
```

Query diatas akan menampilkan daftar nilai maksimum pada kolom tersebut yang disesuaikan sesuai dengan kolom yang digrouping.

Sekarang kita dapat menambahkan di baris berikutnya tentunya dengan menggunakan fungsi Max untuk mendapatkan jumlah total_price yang berada di atas 1.000.000.

```
2 SELECT product_id, MAX(total_price) AS total FROM invoice GROUP BY product_id HAVING  
MAX(total_price) > 1000000;
```

Fungsi Max pada Having

select product_id, max(total_price) as total from invoice group by product_id;

<u>product_id</u>	total
10001	100000
10002	250000
10003	300000
10004	500000
10005	1200000

select product_id, max(total_price) as total from invoice group by product_id having max(total_price) > 1000000;

<u>product_id</u>	total
10005	1200000



Fungsi Min pada Having

```
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1 SELECT product_id, MIN(total_price) AS total FROM invoice GROUP BY  
   product_id;
```

Query diatas akan menampilkan daftar nilai minimum pada kolom tersebut yang disesuaikan sesuai dengan kolom yang di grouping.

Sekarang kita tambahkan fungsi Min di Having untuk memfilter nilai minimum total_price yang dibawah 500.000.

```
2 SELECT product_id, MIN(total_price) AS total FROM invoice GROUP BY  
   product_id HAVING MIN(total_price) < 500000;
```


Fungsi Min pada Having

select product_id, min(total_price) as total from invoice group by product_id;

<u>product_id</u>	<u>total</u>
10001	100000
10002	250000
10003	300000
10004	500000
10005	1200000

select product_id, min(total_price) as total from invoice group by product_id having min(total_price) < 500000;

<u>product_id</u>	<u>total</u>
10001	100000
10002	250000
10003	300000

Fungsi AVG pada Having

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```
1 SELECT product_id, AVG(total_price) AS total FROM invoice GROUP BY  
   product_id;
```

Dari query tersebut mendapatkan list nilai tagihan rata-rata setiap product_id. Sekarang menggunakan having untuk memfilter nilai rata-rata dari total_price.

```
2 SELECT product_id, AVG(total_price) AS total FROM invoice GROUP BY  
   product_id HAVING AVG(total_price) > 100000;
```

Fungsi AVG pada Having

```
select product_id, avg(total_price) as total from invoice group by product_id;
```

<u>product_id</u>	total
10001	100000
10002	250000
10003	300000
10004	500000
10005	1200000

```
select product_id, avg(total_price) as total from invoice group by product_id having avg(total_price) > 100000;
```

<u>product_id</u>	total
10002	250000
10003	300000
10004	500000
10005	1200000

Quiz 1

- Buat query pada tabel **orderdetails** untuk menampilkan kolom **ordernumber**, **itemscount** (jumlah quantityordered) dan **total** (total penjualan/sales)
- Gunakan **GROUP BY** untuk mengelompokkan ordernumber

	ordernumber	itemsCount	total
▶	10100	151	10223.83
	10101	142	10549.01
	10102	80	5494.78
	10103	541	50218.95
	10104	443	40206.20
	10105	545	53959.21
	10106	675	52151.81
	10107	229	22292.62

Quiz 1

- Setelah itu buat query lagi untuk **menampilkan total sales yg lebih besar dari 1000** menggunakan **HAVING**

	ordernumber	itemsCount	total
▶	10100	151	10223.83
	10101	142	10549.01
	10102	80	5494.78
	10103	541	50218.95
	10104	443	40206.20
	10105	545	53959.21
	10106	675	52151.81
	10107	229	22292.62

Quiz 1

- Setelah itu buat query lagi untuk selain **menampilkan total sales yg lebih besar dari 1000** juga menampilkan **itemscount yg lebih besar dari 600**

	ordernumber	itemsCount	total
▶	10106	675	52151.81
	10126	617	57131.92
	10135	607	55601.84
	10165	670	67392.85
	10168	642	50743.65
	10204	619	58793.53
	10207	615	59265.14
	10212	612	59830.55
	10222	717	56822.65

Quiz 2

- Buat query join antara tabel **orderdetails** dengan **order** yang menampilkan **status shipped** dan **total sales yang lebih besar dari 1500**

	ordernumber	status	total
▶	10100	Shipped	10223.83
	10101	Shipped	10549.01
	10102	Shipped	5494.78
	10103	Shipped	50218.95
	10104	Shipped	40206.20
	10105	Shipped	53959.21
	10106	Shipped	52151.81

Terima Kasih